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Lesion recovery of two scleractinian corals under low pH conditions: Implications for restoration efforts

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ABSTRACT

Some coral restoration efforts are involving cultivation of coral microfragments in land-based pools under controlled conditions until they reach viable size for outplanting. However, gaps in knowledge with these efforts include effects of changing pH on regeneration rates of tissue lesions and other physiological responses on different size fragments. To address this, two fragment sizes of *Porites porites* and *Porites astreoides* were artificially inflicted with lesions and incubated in two pH treatments to follow effects on recovery and physiological performance. Recovery was significantly reduced at reduced pH for *P. porites* in both fragment sizes; while recovery of *P. astreoides* was reduced only in the larger fragments. Different responses were also seen for *Symbiodinium* density and total protein with pH and fragment size. Effects on lesion recovery rate from pH and fragment size were species specific and may be related to morphology and/or energetic constrains.

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1. Introduction

Worldwide, coral reef degradation, caused by natural and anthropogenic stressors has been widely reported (Hughes et al., 2003; Grimsditch and Salm, 2006; Hoegh-Guldberg and Bruno, 2010). The Florida Keys reef ecosystem has already experienced long-term deterioration from environmental and anthropogenic stressors and restoration efforts are currently taking place (Lirman et al., 2010; Johnson et al., 2011; Page and Vaughan, 2013). Efforts have shown great promise (Johnson et al., 2011); however it is important to refine restoration techniques in light of future climate change. Previous approaches for restoration efforts include relying on natural recruitment (Edwards and Clark, 1999) yet this process can take several years and is sometimes unreliable when a reef is already damaged. Another approach has been transplantation of coral fragments (Hudson and Diaz, 1988; Johnson et al., 2011; Page and Vaughan, 2013). Transplantation of fragments has shifted from using broken or rescued coral fragments to growing coral fragments in land and field-based nurseries. New techniques are even involving microfragments grown in land-based raceways under controlled conditions until they reach viable size for transplantation onto the reef where they may grow faster and fuse together to form larger colonies (Page and Vaughan, 2013). However, there are gaps in knowledge that need to be addressed through experimental research to support these restoration efforts including effects of future climate scenarios on the regeneration rate of lesions on different size fragments used for restoration purposes. Evidence has shown that scleractinian corals, key reef builders, may be greatly affected by decreased oceanic pH (Hoegh-Guldberg et al., 2007; Jokiel, 2011). The existence of coral reefs depends heavily on the skeletal expansion and changes in density of scleractinian corals which is threatened by ocean acidification (Fabricius et al., 2013). The rate of skeletal growth of corals is not only essential to coral fitness, reproductive output, and ecological success, but also for the ability to repair damage (Ferrari et al., 2012; Tanzil et al., 2013). Corals are frequently subjected to a multitude of disturbances that cause lesions including: algal abrasions or overgrowth, storm damage, sedimentation, disease, tourism (e.g. physical damage by breaking off pieces of coral for collection, lesions from careless snorkeling or diving, anchor damage, and boat groundings), and fish bites (Jayewardene, 2010; Palmer et al., 2011; Bulleri et al., 2013; White et al., 2013; Hamilton et al., 2014). Coral cultivation for restoration purposes also inflicts lesions when a coral colony is fragmented. Regeneration of tissue across a coral lesion is mainly fueled by the bordering polyps and tissue, and begins with the inward progression of undifferentiated tissue created by the polyps followed by new polyps (Denis et al., 2011). Regeneration is also likely dependent on the lesion surface area to perimeter ratio (Meesters et al., 1994; Jayewardene, 2010), and varies with species (Denis et al., 2013; Horwitz and Fine, 2014). However, regeneration of lesions can represent a significant drain on energy resources available for coral

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maintenance, growth, and reproduction and can have long lasting effects on coral colonies (Guzmán et al., 1994).

Current goals for coral reef restoration include identifying, understanding, and communicating risks of corals to climate change, ocean acidification, and other environmental perturbations as well as enhancing strategic management of coral reef ecosystems through improved and applied understanding of those impacts (NOAA, 2009). Therefore, it is important to understand how corals will be affected in future climate scenarios. Preparation of fragments for restoration includes fragmentation of hundreds of corals, resulting in lesions that have to recuperate quickly in order to prevent pathogens and boring organisms from penetrating to the coral tissue and exposed skeleton (Bowden-Kerby, 2001; Rinkevich, 2005). Faster healing processes will likely allow for a better chance of survival and ability to allocate more energy to growth. Nevertheless, once those fragments are placed in the wild, they are just as susceptible to new lesions. Therefore, it is also important to understand how different restoration efforts and techniques will be affected in future climate scenarios, especially changes in pH (or ocean acidification). Ocean acidification refers not only to changes in seawater pH but also changes in other carbonate chemistry parameters, however changes in pH alone are potentially enough of a stressor based on the "proton flux hypotheses" (Jokiel, 2011). Here, we report on how reduced pH conditions affect lesion regeneration in varying sized fragments of *Porites porites* and *Porites astreoides*. We chose to use future predicted values of sea surface pH values for the next century (IPCC, 2014). This study is the first to determine if fragment size effects tissue regeneration in low pH conditions and could be greatly beneficial to restorative efforts.

2. Materials and methods

2.1. Experimental design

Four colonies of two western Atlantic corals (*P. porites* and *P. astreoides*) were collected from Wonderland Reef (24°33.69 N, 081°30.18 W) located in the Lower Keys portion of the Florida Keys National Marine Sanctuary in August 2013 at 8 m depth and transported by boat to the Mote Tropical Research Laboratory (Mote TRL) in Summerland Key, Florida. The colonies were labeled by colony, and allowed to acclimate in a running seawater system (maintained at pH 8.1 and constant temperature of 26 °C) for 24 h before they were divided to experimental aquaria and remained in the system for 60 days at two different pH levels (8.1 and 7.6) prior to fragmentation and lesion infliction. Seawater pH was adjusted by aeration using the well-water system as described in Hall et al. (2012). Water turnover was adjusted to 50% aquarium volume per hour. The pH, temperature, and salinity of individual aquarium tanks were monitored twice daily using a Mettler Toledo pH meter (SevenGo proTM, OH, USA) with automatic temperature compensation and pH combination electrode (InLab, Mettler Toledo, OH, USA) and a YSI multiprobe (Pro1030, OH, USA). Prior to experiments, the pH electrode was calibrated against National Bureau of Standards (NBS) scale buffers of 4.01, 7.00, and 10.00 at 25 °C. The manufacturer's technical specifications of the pH meter are 0.01 pH for resolution and ±0.01 (standard error) for accuracy. After 60 days in treatment tanks, corals were sectioned into seven 1 cm^2 (1 × 1) and seven 25 cm^2 (5 × 5) fragments and allowed to acclimate again for 30 days. After acclimation, artificial circular lesions (10% of surface area) were inflicted on coral fragments with an air brush and fragments were separated into 12 aquaria with 4–5 replications of each pH and fragment size combination: 8.1/1 × 1; 8.1/5 × 5; 7.6/1 × 1; and 7.6/5 × 5 in flow through aquarium tanks. Fragments were allowed to acclimate for 10 more days to allow for healing. For lesion recovery measurements, photographs were taken of each coral by digital camera at a fixed distance and processed with ImageJ® software daily (Fig. 1). Recovery of lesions was quantified and expressed as percent tissue regeneration from day 1 to day 20. After 20 days, short term physiological and functional responses were measured using chlorophyll *a*, total protein, zooxanthellae counts, and surface area as described below. Short term responses were measured because many of the fragments were fully recovered from lesions after 20 days.

2.2. Tissue removal, surface area, zooxanthellae, chlorophyll a, and protein

At the end of the experiment, coral tissue was completely removed from each fragment using an air brush and filtered seawater (0.2 μm FSW). The skeleton was retained for surface area determination following methods by Marsh (1970). Briefly, pieces of aluminum foil were wrapped around the coral skeleton to cover all surface area of the tissue

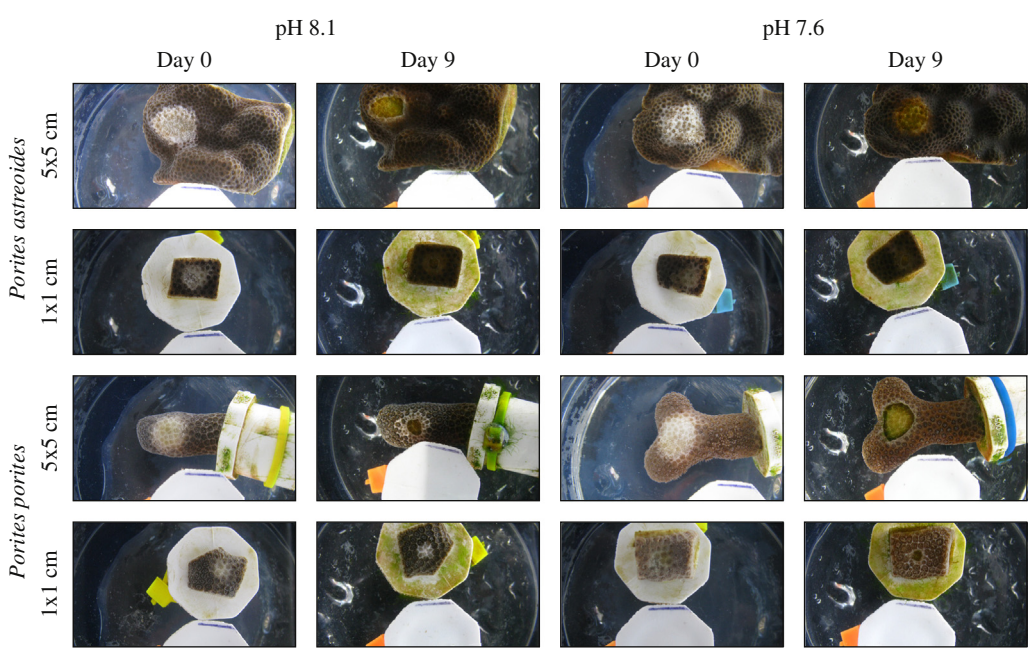

Fig. 1. Lesion recovery of representative *Porites porites* and *P. astreoides* after 10 days under 2 pH conditions (C-8.1 and L-7.6) and for 2 fragment sizes (1 × 1 cm and 5 × 5 cm).

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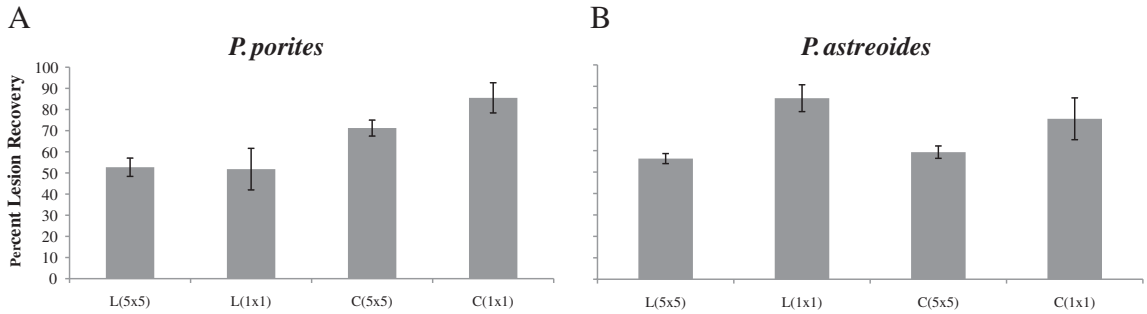

Fig. 2. Average percent lesion recovery (mean ± SEM, n = 9) after 10 days for (A) *Porites porites* and (B) *P. astreoides* in low (L-7.6) and control (C-8.1) pH and two fragment sizes (1 × 1 cm and 5 × 5 cm).

(excluding the lesion area). The aluminum foil was then removed, measured, and weighed with an analytical balance and calibrated against a standard linear curve for surface area and weight. To estimate zooxanthellae density and chlorophyll concentration, the tissue slurry was collected in a Falcon tube with up to 45 mL of the sample slurry and homogenized. A sub-sample of the slurry (1 mL) was taken for zooxanthellae counts. Symbiont densities were counted using a compound light microscope (Leica CME, Buffalo, Illinois, USA) and hemocytometer. The remaining tissue slurry for each colony was centrifuged at 5000 rpm for 5 min. Subsequently, the supernatant was removed and the zooxanthellae were re-suspended in 5 mL of 90% acetone for extraction of chlorophyll *a*. Samples were kept in darkness at 4 °C for a period of 24 h to ensure total extraction. Chlorophyll content was determined using a spectrophotometer (Perkin Elmer, Massachusetts, USA) based on methods by (Jeffrey and Humphry, 1975). Samples were measured at 750, 663 and 630 nm. Absorbance at 750 nm was used to control for sample turbidity, and chlorophyll content was calculated from absorbance values at 663 and 630. A 100 μL subsample of the slurry was taken for total protein analysis using the Bradford protocol (Bradford, 1976). Samples were mixed with dye reagent (200 μL, Coomassie Brilliant Blue G-250) and incubated for 30 min in the dark at room temperature before the optical density was read on a spectrophotometer (595 nm, Eppendorf BioPhotometer plus, Germany). Samples were run against a standard calibration curve of Bovine Serum Albumin and were normalized to surface area.

2.3. Statistics

The software package Systat11® was used to examine independent and interaction relationships between treatment variables. One-way ANOVAs (with pH and fragment size as the independent variables) were run on percent recovery rates, zooxanthellae density, chlorophyll *a*, and protein for *P. porites* and *P. astreoides* to compare pH (L and C) and fragment size (5 × 5 cm and 1 × 1 cm) treatments. Kruskal–Wallis one tests were run for non-normal sample sets. Shapiro–Wilk tests were run for normality.

3. Results

3.1. Lesion recovery

Lesion recovery rate was significantly lower at pH 7.6 than pH 8.1 (average 78% compared to average 52% after 9 days of incubation) for both fragment sizes of *P. porites* ($F_{(1,32)} = 15.087$, $p < 0.001$, Fig. 2(a)). No significant differences were found between fragment sizes for *P. porites* at either pH as determined by one-way ANOVA. In *P. astreoides* however, no significant difference in recovery rate was found between pH treatments however fragment size showed significant reduction in percent lesion recovery rate for larger fragments as determined by Kruskal–Wallis ($p < 0.05$, chi-squared approx. = 7.114 with 1 df, Fig. 2(b)).

3.2. Total protein, Symbiodinium density, and chlorophyll concentration

Exposure to reduced pH showed no significant effect on *Symbiodinium* density of *P. porites*. However, *Symbiodinium* density was reduced in larger fragment sizes ($F_{(1,24)} = 8.261$, $p < 0.05$, Fig. 3(a)). Similarly, *Symbiodinium* density of *P. asteroides* was also only reduced with the larger fragment size ($F_{(1,24)} = 25.434$, $p < 0.001$, Fig. 3(b)). No significant differences in chlorophyll *a* concentration were seen in either species with any treatment (Fig. 4). Total protein concentration per surface area in *P. porites* was significantly lower with reduced pH treatment in both fragment sizes ($F_{(1,24)} = 20.254$, $p < 0.001$, Fig. 5(a)). In *P. asteroides* total protein concentration per surface area was also significantly lower with reduced pH treatment ($F_{(1,24)} = 17.166$, $p < 0.001$) and was significantly lower in larger fragments than in smaller fragments ($F_{(1,24)} = 10.460$, $p < 0.05$, Fig. 5).

4. Discussion

This study investigated the effect of reduced pH and fragment size on lesion recovery rates of con-generic coral species used for coral reef restoration. Lesion recovery is a life-preserving process that often

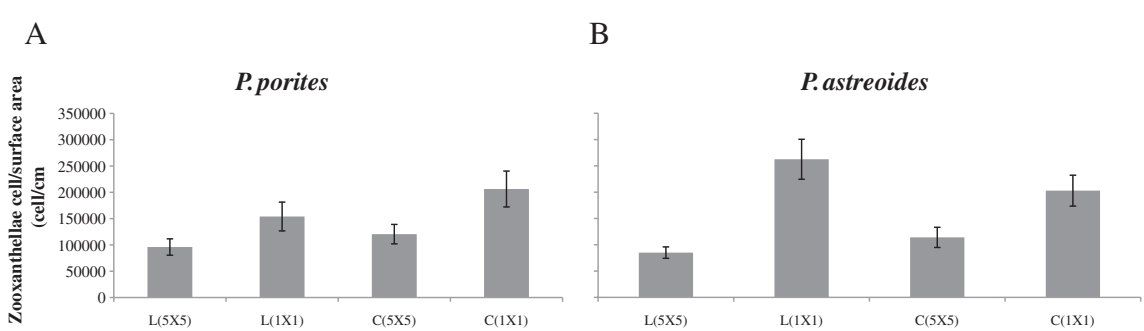

Fig. 3. Zooxanthellae cell/surface area (mean ± SEM, n = 7) for (A) *Porites porites* and (B) *P. astreoides* in low (L-7.6) and control (C-8.1) pH and two fragment sizes (1 × 1 cm and 5 × 5 cm).

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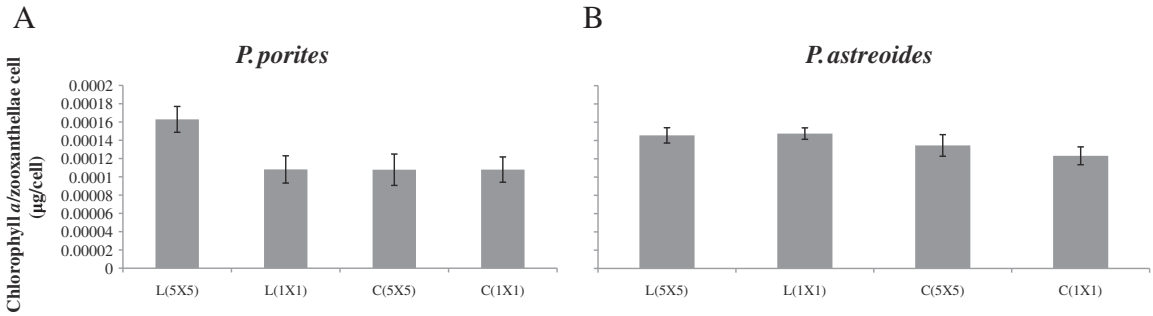

Fig. 4. Chlorophyll *a*/zooxanthellae cell (mean ± SEM, n = 7) for (A) *Porites porites* and (B) *P. astreoides* in low (L-7.6) and control (C-8.1) pH and two fragment sizes (1 × 1 cm and 5 × 5 cm).

derives energy and cells at the expense of other physiological processes (Meesters et al., 1997). Lesion recovery rates have been extensively studied (Lirman, 2000; Cròquer et al., 2002; Henry and Hart, 2005; Tentori and Allemand, 2006; Titlyanov and Titlyanov, 2009), however many studies have been long-term (>30 days) and haven't focused on restoration efforts. In this short term study, we saw 90–100% recovery in 22% of individual fragments, all of which were from the smaller fragment size in both species. Our data also agrees with others (i.e. Horwitz and Fine 2014) that lesion recovery rates are reduced in low pH conditions and with fragment size, however these effects are species-specific. Jokiel (2011) found that reduced pH in the seawater surrounding corals limits removal of [H+] from the site of calcification in the corals and from coral cells which could potentially depress intracellular pH and thus compromise the capacity of cells to regenerate and repair lesions (Fig. 5).

Other studies have determined that lesion recovery is influenced by size and shape of the lesion. Lirman (2000) found that lesion regeneration in *Acropora palmata* is influenced by the initial size and perimeter of the lesions, but not by the size of colonies or fragments bearing the lesions. Oren et al. (1997) also found that both size and shape of lesions influence recovery rates. In the present study, each fragment was inflicted with the same percent lesion area (10% of the fragment size). Therefore, lesion area was reduced with fragment size and our results agree with other studies showing that larger lesions do not show as significant recovery over time. Location of lesions on coral colonies or fragments might also be a factor in recovery rate. If lesions are nearer younger polyps where energy is being used for faster growth, then recovery rates may be faster (Pearse and Muscatine, 1971). The role of lesion position on fragments was not examined during this study but may have an effect, especially on the larger, branching colonies.

The effects of colony or fragment size on lesion recovery have been studied previously with differing results (Loya, 1976; Bak and Van Es, 1980; Wahle, 1983; Chadwick and Loya, 1990; Andres and Rodenhouse, 1993; Lirman, 2003). Some coral species have shown reduced recovery capabilities in smaller colonies (Loya, 1976), others show that small solitary colonies recover faster than large corals with similar amounts of damage (Chadwick and Loya, 1990; Andres and Rodenhouse, 1993; Lirman, 2000), and others have shown no relationship between colony size and regeneration rate (Wahle, 1983). After damage, repair requires metabolic resources in the form of carbon which are trans-located from undamaged portions of the coral (Cameron and Edmunds, 2014). Thus, even when corals of varying sizes' tissues are damaged to the same percentage, the larger coral should be able to recover faster due to its ability to translocate resources from larger tissue mass faster. It has also been shown that expenditure of energy for lesion recovery is lower in larger sizes or that larger colonies have more energy available for regeneration (Loya, 1976). A larger size fragment can enhance stability as new corals are outplanted or during storm events and serve as a larger resource area for lesion regeneration (Highsmith et al., 1980; Bak 1983). However, the threshold size needed for complete regeneration will depend on the degree of injury and is likely species-specific (Bak 1983). While size of fragment did not affect recovery rate in *P. porites*, this was not the case with *P. astreoides*, indicating that fragment size and rate of lesion recovery is species-specific.

Differences between con-generic species may be an effect of energy allocation capabilities that are species specific or due to different morphologies. *P. porites* and *P. astreoides* are con-generic corals that have previously shown differential responses to different stressors (Hall et al. unpublished). In the present study, only *P. astreoides* showed reduced lesion recovery rates in larger fragments, regardless of pH while *P. porites* only showed reduced recovery under reduced pH. *P. porites* is a branching coral which often has thinner tissue layers that provides less energy-rich compounds, resulting in lower resistance to environmental changes (Jiménez et al., 2001; Loya et al., 2001; McClanahan et al., 2001). Independent of morphology, regulation and compensation of the organism intra- and extracellular acid–base balance is energy costly and even more so under a high pCO_2/low pH conditions when compensation is required to maintain homeostasis (Fabry et al. 2008). Under stressful conditions (such as acidosis) the coral resources are re-directed into colony maintenance and cell pH

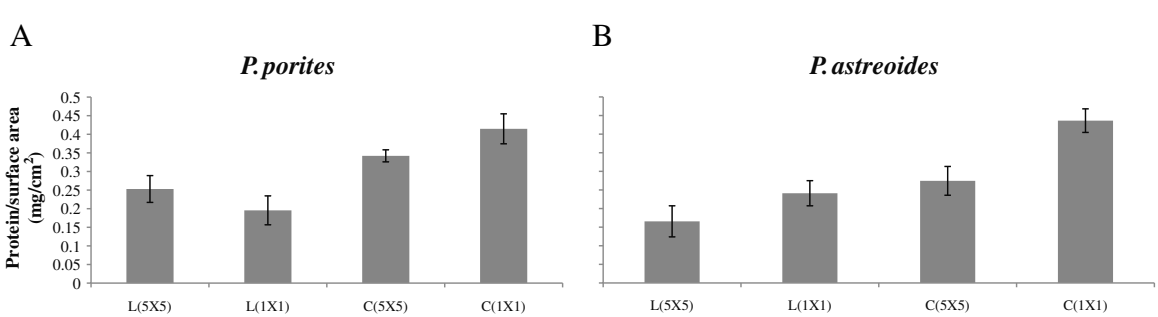

Fig. 5. Protein/surface area (mean ± SEM, n = 7) for (A) *Porites porites* and (B) *P. astreoides* in low (L-7.6) and control (C-8.1) pH and two fragment sizes (1 × 1 cm and 5 × 5 cm).

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homeostasis, at the cost of metabolic depression and changes in cell membrane transporters (Pörtner et al., 2000; Kaniewska et al., 2012). Intra and extracellular pH in corals was shown to be regulated and maintained at a steady state under increased pCO_2 conditions (Venn et al., 2013), suggesting increased energy investment. For an energetically costly process such as tissue regeneration, allocation of energy towards basic cellular homeostasis (acid–base regulation) may mean energetic deficiency, leading to slower regeneration.

Recovery time might also be a factor in lesion recovery rates. Recovery of a damaged surface slows down with time, and may result in permanent lesions, although some corals such as *A. palmata*, can fully recover from all lesions in 80 days (Bak 1983). *Montastraea annularis* showed an exponential decrease in recovery rate as time increased and is also influenced by size, type, and position of the lesion (Cròquer et al., 2002). In the present study, only small fragments of both species showed 100% recovery within 9 days. Of the completely recovered fragments, the majority for *P. porites* were in control pH treatment while there were no differences with pH for *P. astreoides*. Regeneration of lesions also requires reallocation of energy such as calcification and protein production (Bak 1983). Previous work has shown that calcification of *Porites* spp. in general is reduced in low pH environments (Lough and Barnes, 2000; Albright et al., 2008; Cohen et al., 2009). The ability of corals to repair damage and cope with stressors can also be attributed to protein concentration. In the present study, protein concentration was reduced in low pH environments for both species. Higher protein concentration per surface area in smaller fragments of *P. astreoides* may explain the faster regeneration rate in small fragments compared to larger fragments. Lower concentrations of *Symbiodinium* are an indicator of stress as well (Tremblay et al., 2013; Horwitz and Fine 2013). In this study, *P. astreoides* had lower symbiont density in the larger fragment size than the smaller size, corresponding with lower recovery rates.

Our data agrees with others, showing higher recovery rates in ambient pH conditions, however there are many other factors not taken into account in this study such as temperature (Meesters and Bak, 1993; Kramarsky-Winter and Loya, 2000), bleaching (Meesters and Bak, 1993; Fine et al., 2002), food availability (Fisher et al., 2007), sedimentation (Meesters et al., 1992), continuous grazing, and other environmental impacts. We found that the two studied species are able to function and recover from injury in different environmental conditions. This is likely a consequence of shifting of energy from growth, calcification, reproduction, or other functions. We also show that fragment size and seawater pH do have implications for lesion regeneration. Although *P. porites* and *P. astreoides* are two common coral species that are prevalent in the western Atlantic and are not currently a main focus for restoration efforts, environmental changes have reduced percent coverage of corals worldwide, and *P. astreoides* and *P. porites* have become a focus for environmental effects on what's considered a more resilient coral (Lirman 2003; Edmunds and Gates, 2008; Aronson et al., 2008; Albright et al., 2008; Green et al., 2008; Albright and Langdon, 2011; Crook et al., 2013). In the short term, small fragment size might be more beneficial for lesion recovery, although this is species-specific. pH also has an influence on lesion recovery rate and must be taken into account as oceanic chemical shifts are occurring (IPCC, 2014). Energy shifts in corals is vital for lesion recovery, but is also species specific. We feel that this is beneficial information for restoration efforts; however there are other factors that should be examined. There may be greater risk of prey of an entire colony of smaller fragments if not protected during first few months of growth when outplanted and recovery rates may slow down over longer periods of time.

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